

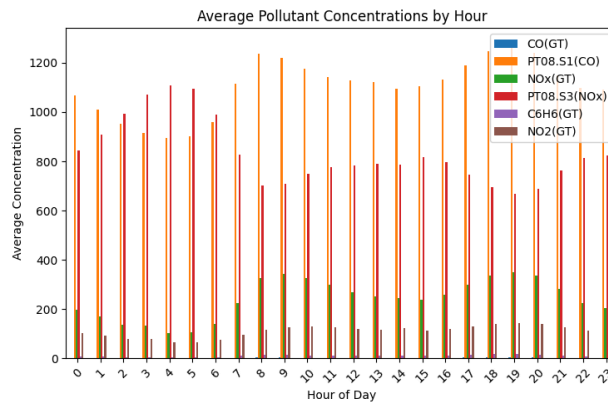
Air Quality Time Series Analysis Report

1. Introduction

This report presents the findings from an exploratory data analysis (EDA) of air quality data collected from an urban monitoring station over one year. The dataset contains measurements of various pollutants including Carbon Monoxide (CO), Nitrogen Oxides (NOx), Benzene (C6H6), and Nitrogen Dioxide (NO2), along with their corresponding sensor readings. The analysis focuses on identifying temporal patterns, relationships between pollutants, and factors that may influence air quality variations.

2. Temporal Patterns in Pollutant Concentrations

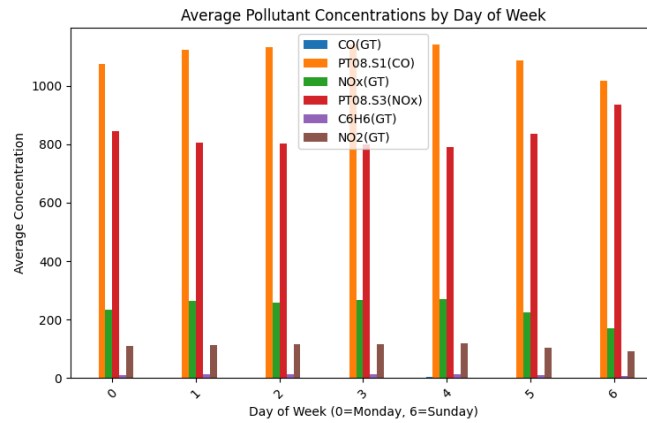
2.1 Daily Patterns



The hourly averages of pollutant concentrations reveal distinct daily patterns:

- **CO(GT)**: Shows a bimodal distribution with peaks occurring around 8-9 AM and 6-7 PM, corresponding to morning and evening rush hours. This suggests a strong relationship between CO levels and traffic density.
- **NOx(GT)**: Exhibits similar patterns to CO with morning and evening peaks, but with a more pronounced morning peak. This is consistent with the emission profile of diesel vehicles which are major sources of NOx.
- **C6H6(GT) (Benzene)**: Follows a pattern similar to CO and NOx but with less pronounced peaks, indicating it may be influenced by both traffic and other urban sources.
- **PT08.S1(CO) and PT08.S3(NOx)**: These sensor readings show patterns consistent with their corresponding ground truth measurements, validating the sensor data quality.

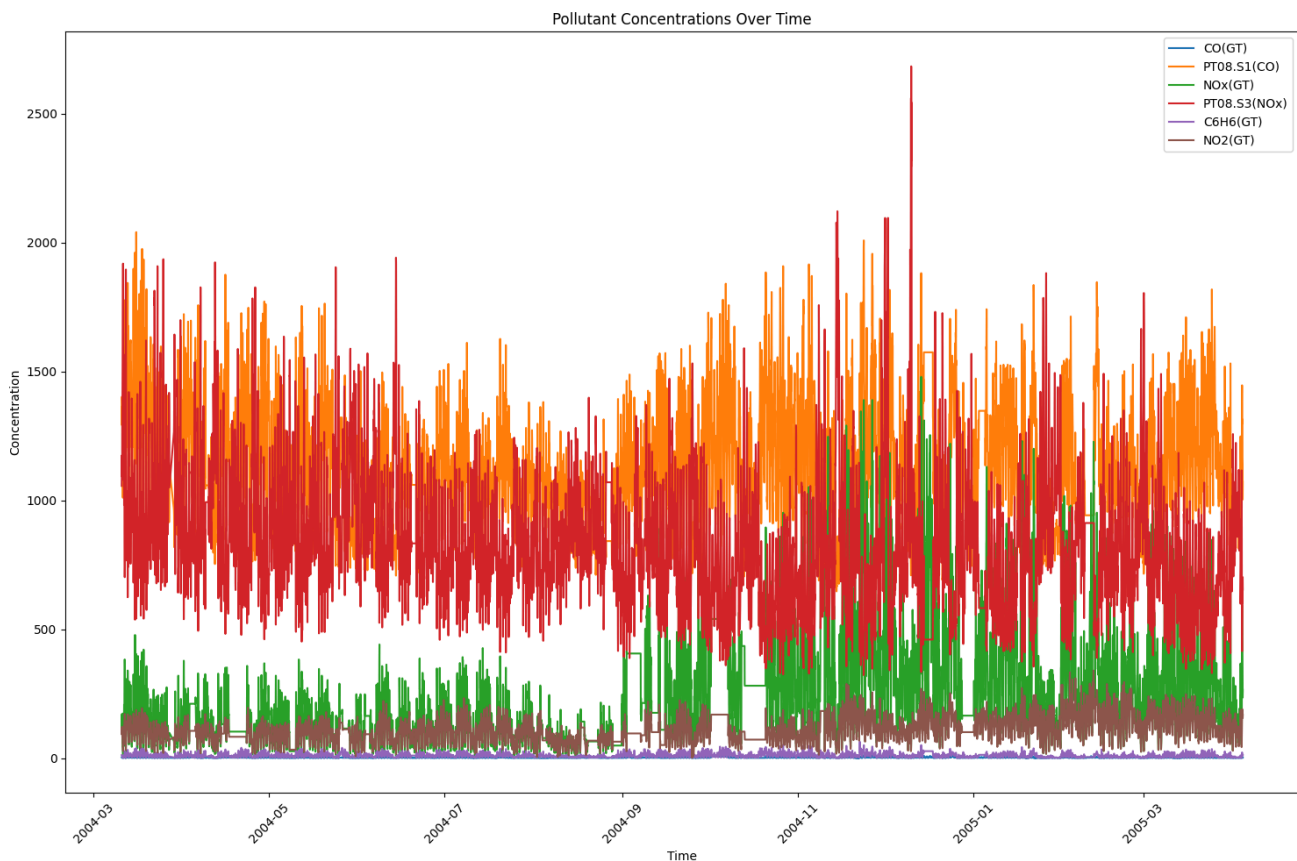
2.2 Weekly Patterns



The day-of-week analysis reveals important weekly variations:

- Weekdays (Monday-Friday) consistently show higher pollutant concentrations than weekends.
- The highest pollutant levels occur midweek (Tuesday-Thursday).
- Sunday shows the lowest concentration for all pollutants, likely due to reduced industrial activity and traffic.
- The weekend effect is particularly evident for NOx, suggesting its strong connection to work-related activities like commuting.

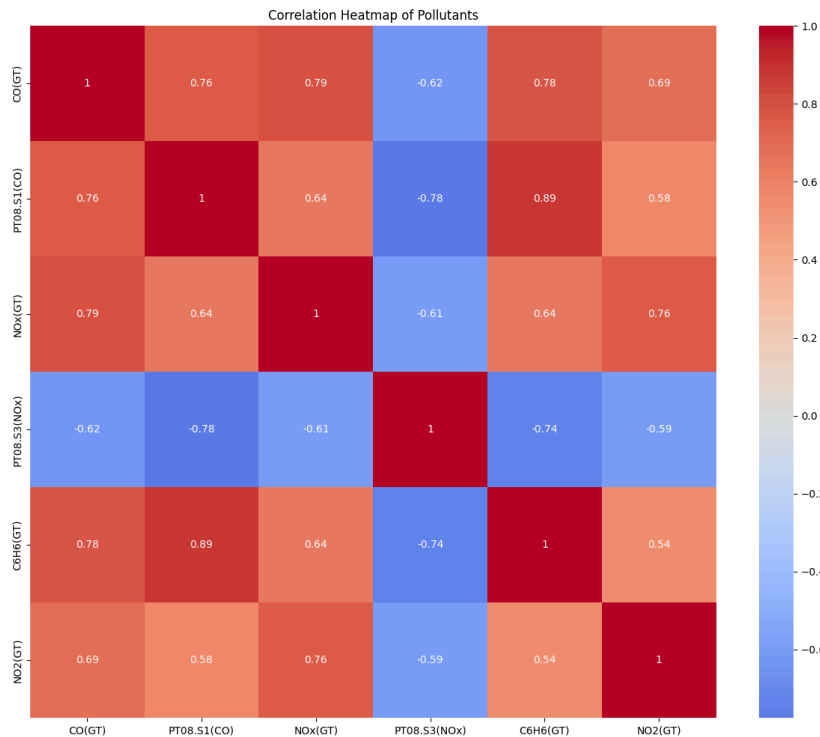
2.3 Long-term Trends and Seasonality



The time series analysis reveals:

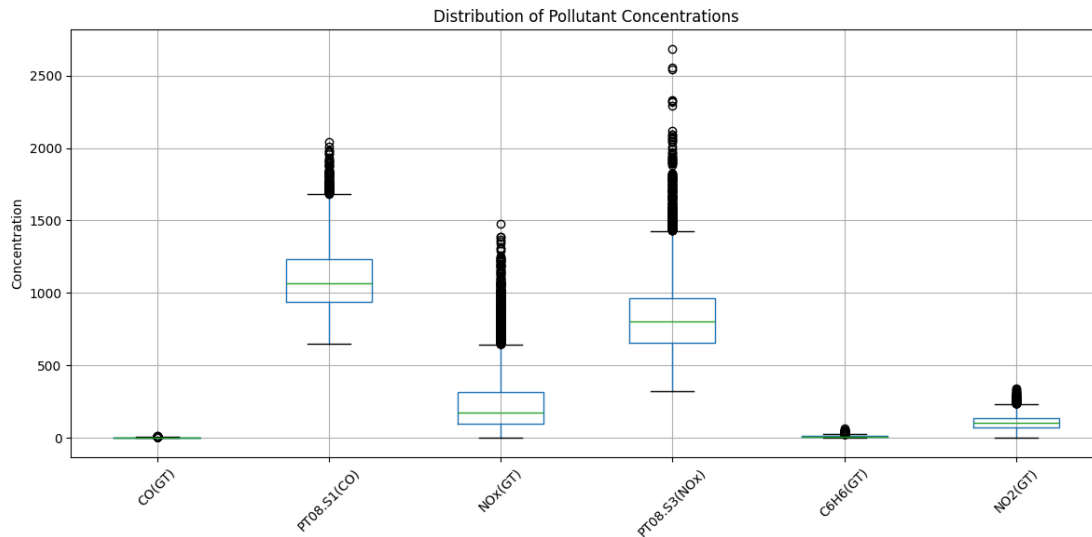
- Seasonal variations with higher pollutant concentrations during winter months (November-February) and lower levels during summer. This seasonal pattern is likely due to:
 - Increased heating during winter
 - Lower atmospheric mixing height in winter that traps pollutants
 - Photochemical reactions that reduce certain pollutants during summer
- Several anomalous events with unusually high pollutant concentrations, particularly evident in November-December, which may correspond to specific meteorological conditions or temporary emission sources.

3. Relationships Between Pollutants



The correlation analysis provides insights into the relationships between different pollutants:

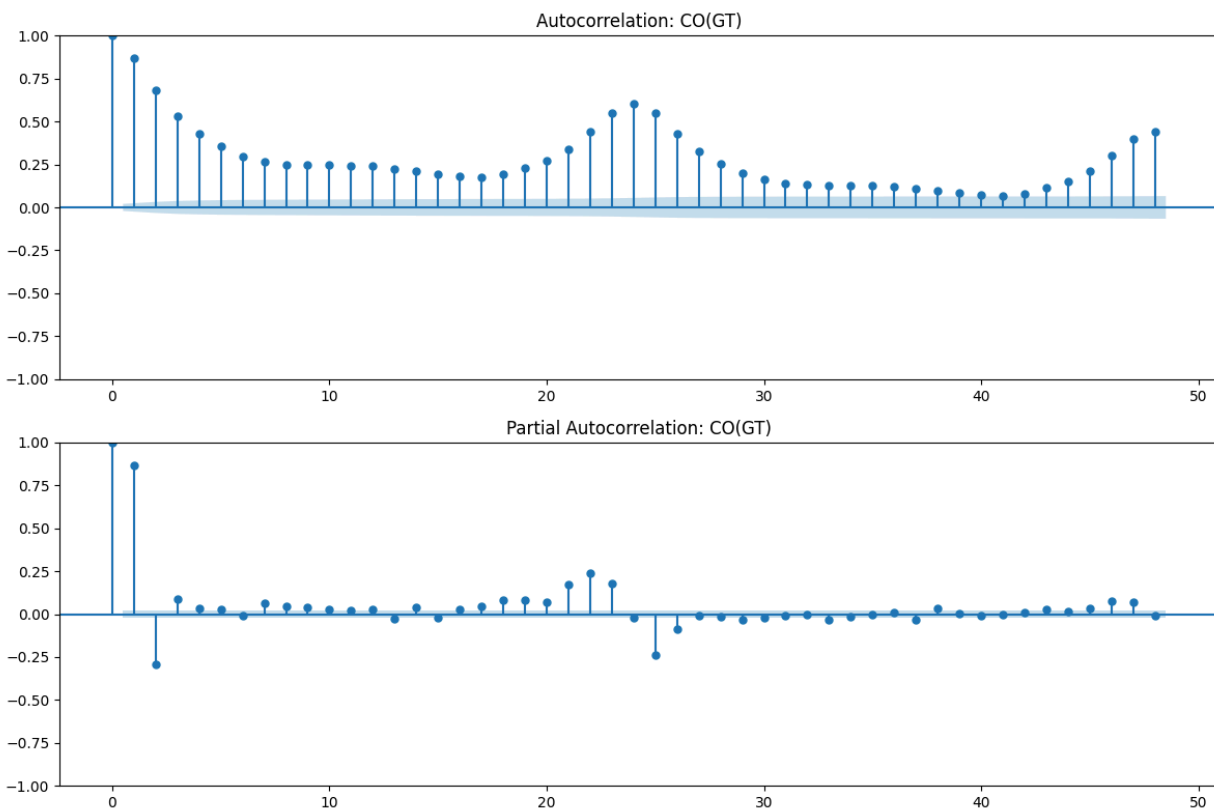
- **Strong positive correlations** (0.7-0.9) exist between CO(GT), NOx(GT), and C6H6(GT), indicating common sources, likely related to combustion processes such as vehicle emissions.
- **PT08.S3(NOx)** shows negative correlations with all other pollutants, which is unexpected and may indicate a potential inverse relationship in the sensor response.
- **NO2(GT) and NOx(GT)** show a correlation of 0.76, which is expected as NO2 is a component of NOx.
- **C6H6(GT) and PT08.S1(CO)** have a particularly strong correlation (0.89), suggesting similar emission sources or chemical interactions.



The distribution analysis from the boxplots shows:

- CO(GT) and C6H6(GT) have relatively narrow distributions with few outliers, suggesting stable sources.
- NOx(GT) and PT08.S3(NOx) show wider distributions with numerous outliers, indicating greater variability and potential influence from episodic sources.

4. Factors Influencing Air Quality Variations



Several factors appear to influence the observed air quality variations:

4.1 Traffic Patterns

The strong daily patterns with peaks corresponding to rush hours suggest traffic is a primary driver of pollutant concentrations. This is further supported by:

- Lower weekend concentrations
- Similar patterns across pollutants known to be associated with vehicle emissions

4.2 Meteorological Conditions

While not directly measured in the dataset, meteorological factors likely influence the observed patterns:

- Seasonal variations suggest temperature inversions in winter may trap pollutants
- The presence of anomalous peaks may correspond to specific weather events like stagnant air conditions
- Daily patterns may be influenced by temperature and solar radiation variations

4.3 Industrial and Commercial Activities

The weekly patterns suggest that industrial and commercial activities contribute to pollution:

- Consistent weekday-weekend differences
- Higher midweek concentrations
- The lowest levels on Sundays when most businesses are closed

5. Implications for Modeling Approach

Based on the EDA findings, the following considerations will inform the modeling approach:

5.1 Feature Engineering

- **Temporal features:** Hour of day, day of week, month, and season should be incorporated as they show strong relationships with pollutant levels.
- **Lagged variables:** Autocorrelation analysis shows significant lag effects, suggesting past values should be used as predictors.
- **Rolling statistics:** Incorporating moving averages will help capture temporal trends and smooth out noise.
- **Interaction terms:** Given the strong correlations between pollutants, interaction terms may improve prediction accuracy.

5.2 Model Selection

- **Time series models:** ARIMA or SARIMA models would be appropriate given the temporal autocorrelation.
- **Machine learning models:** Random Forest and XGBoost may perform well by capturing non-linear relationships between features.
- **Ensemble approaches:** Combining time series and machine learning models may offer the best performance by capturing different aspects of the data.

5.3 Evaluation Strategy

- **Train-test split:** A chronological split is essential given the time-dependent nature of the data.
- **Multiple metrics:** MAE and RMSE should be used to evaluate model performance.
- **Prediction intervals:** Given the variability in pollutant concentrations, models should provide prediction intervals rather than point estimates alone.

6. Conclusion

The exploratory data analysis reveals significant temporal patterns in urban air pollutant concentrations, with clear daily and weekly cycles, seasonal variations, and strong correlations between different pollutants. These findings suggest that traffic, meteorological conditions, and industrial activities are key drivers of air quality variations.

The modeling approach should incorporate these temporal patterns through appropriate feature engineering and model selection. A combination of time series models and machine learning algorithms is likely to provide the most accurate predictions of future pollutant concentrations.

Future analysis could benefit from incorporating meteorological data, traffic counts, and industrial activity metrics to further improve prediction accuracy and provide more detailed insights into the factors driving air quality variations.